

Realization Limits of the Bicomplex Signal Surface: Analytic Rigidity, Smooth Flexibility and Observation Monodromy

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Abstract

A prescribed power distribution on a two-component signal fixes the moduli of the components but leaves their phases free. We ask whether this freedom can realize an explicit ruled surface $S \subset \mathbb{R}^3$ through a fixed readout map. For the original assignment $(|z_1|^2, |z_2|^2) = (\cos t, 1 - \cos t)$ the answer reveals a sharp distinction between analytic and smooth realizations. No readout that is real-analytic along the circle $\{z_1 = 0\}$ realizes S , whatever states are used; this excludes affine projections, polynomial maps and every polynomial in a bicomplex variable and its conjugates. Explicit C^∞ readouts, however, do exist. The obstruction sits at a single geometric feature: the two edges of the source are forced onto one circle of states, while S maps them onto two different lines. Removing that feature restores analyticity. Each half of the source has a real-analytic readout, and a modified power assignment has a quadratic one.

We also identify the carriers of the square-root monodromy of the observation field. It lives on a local system that is indecomposable over $\mathbb{Z}$ and splits after inverting 2, and it is distinct from both the fold involution and the normalization defect. The exceptional pullback of $\mathbb{Z}$ along any continuous planar map from a convex domain depends only on the domain; folds are seen instead by the comparison map that defines cohomological smoothness. This paper supersedes an earlier preprint, whose named statements are reviewed in an appendix.

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1 Introduction

A prescribed power distribution leaves considerable freedom in the phases. We ask whether this freedom can realize a given surface through a fixed readout. The surface is the explicit ruled map

$$S(t, u) = (\cos t + 2u(1 - \cos t), (1 - u) \sin t, u\sqrt{\cos t(2 - \cos t)}),$$

which the preprint [1] derived from a signal-theoretic picture: a power-conserving two-component state, its moment map, and a projection onto three observed coordinates. Its geometry is rich (Figure 1). It has five cross-cap singularities, an observation field with a topological dipole, and an explicit A_3 residual, and much of this has since been established rigorously [2, 3]. The realization question itself, *which states, read out by which maps, actually produce S ?*, is answered here. The answer reveals a sharp distinction between analytic rigidity and smooth flexibility.

Main results.

1. *Analytic rigidity* (Theorem 3.6). Under the original assignment $|z_1|^2 = \cos t$, every state over the edges $t = \pm\pi/2$ lies on the circle $C_0 = \{z_1 = 0\}$, while S sends the two edges onto a “V” of two non-collinear segments (Figure 2). A real-analytic image of a circle cannot contain such a V. Hence no readout that is real-analytic along C_0 realizes S on the closed source. On the source open in t the same holds for every readout that is real-analytic along C_0 and continuous at the points of C_0 .
2. *Smooth flexibility and sharpness.*
 - An explicit C^∞ readout realizes S (Theorem 3.8).
 - On either half of the source, real-analytic readouts exist, while affine ones still do not (Proposition 3.10).
 - A different power assignment admits a quadratic readout (Theorem 3.2).
 - Readouts that ignore the global phase fail with no regularity hypothesis at all (Proposition 3.4).

An edge criterion (Remark 3.11) explains all of these.

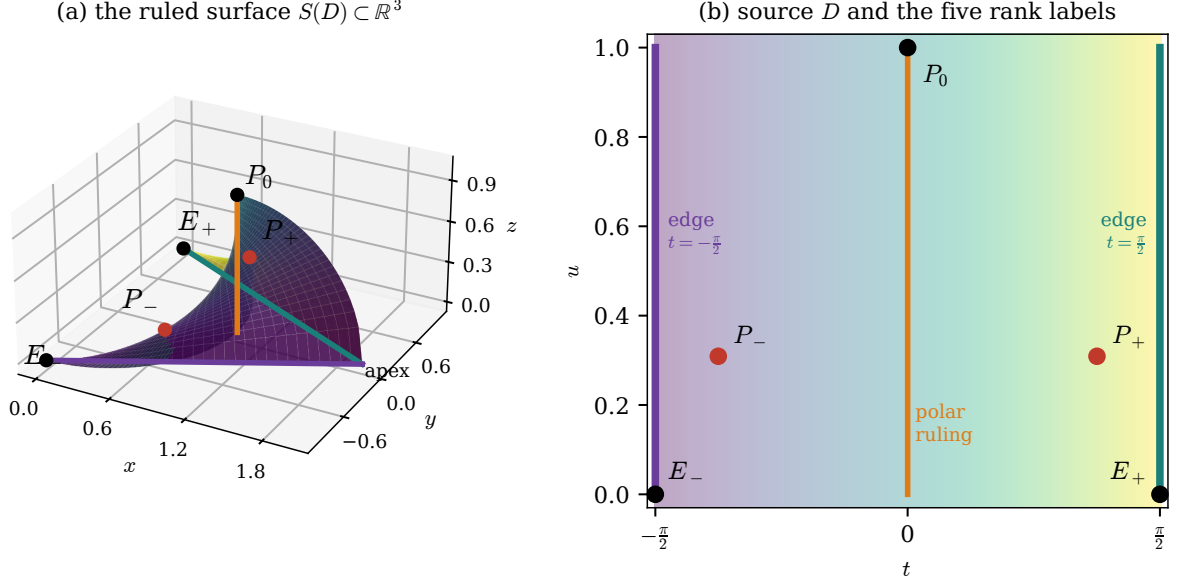


Figure 1: (a) The ruled surface S , coloured by t . The two edges $t = \pm\pi/2$ (teal and purple) lie in the plane $z = 0$ and meet at the apex $(2, 0, 0)$; the polar ruling $t = 0$ (orange) is the vertical segment over $(1, 0)$. (b) The source $D = [-\pi/2, \pi/2] \times [0, 1]$ with the five rank labels of Proposition 2.1: the interior cross-caps $P_{\pm}$, the boundary half-germ P_0 and the corner quarter-germs $E_{\pm}$.

3. *Carriers of the monodromy* (Proposition 4.3). The two-traversal restoration of $\sqrt{F}$ (Figure 4) lives on a rank-two local system that is indecomposable over $\mathbb{Z}$. It differs from the fold involution and from the normalization defect.
4. *Exceptional pullbacks* (Theorem 4.4, Proposition 4.6). For every continuous map from a convex planar domain to the plane, $\Psi^!\mathbb{Z}$ is the extension by zero from the interior. Folds are detected by the smoothness comparison map, whose cokernel is the constant sheaf on the fold arc.

Organization. Section 2 fixes the surface and its observation field. Section 3 proves the rigidity theorem and its sharpness, and Section 4 treats the monodromy and the exceptional pullbacks. Section 5 describes the residual geometry at the cross-caps: a parameter-free 3/4 resolution law, an explicit A_3 completion, and three different projected curves. The earlier versions also contained spectral and arithmetic estimates, for Mellin operators, finite Gram matrices, Hardy evaluation and log-scale windows. They are classical and independent of the realization question, so we collect them, in the precise form the model uses, in Appendix B, together with the scope statements that carry over unchanged (Appendix C).

Relation to the earlier preprint. This paper is the corrected successor of [1]. Appendix A reviews every named statement of its final version and records where each is proved, sharpened or corrected here or in the companion paper [4].

The bicomplex model. We retain “bicomplex” as the name of the model in [1]. Through its idempotent decomposition the bicomplex algebra is $\mathbb{C} \oplus \mathbb{C}$. Writing an element as a pair (a, b) , its three standard conjugations are $(a, b) \mapsto (\bar{a}, \bar{b})$, (b, a) and $(\bar{b}, \bar{a})$, which are generated by componentwise conjugation and the exchange of components. The realization and obstruction theorems below are therefore statements about two-component complex state families; they do

not require bicomplex multiplication as additional structure. Bicomplex algebra enters as a natural class of readouts. Every expression built from bicomplex sums, products, constants and conjugations is a real polynomial map, so this whole class is covered by Corollary 3.7.

Relation to other work. The completed source atlas, the five rank singularities, the dipole indices and the fold classification of the planar observation map were established for the same surface in [2, 3]. We use them as inputs. The Laplace–Beltrami operator of S , its point interactions at the cross-caps, and the Green, Markov and Krein theory that replaces earlier operator statements are developed in the companion paper [4], for general surfaces with Whitney cross-caps. The tools used here are standard: real-analytic functions, smooth partitions of unity, covering spaces, Verdier duality and mixed Hodge theory. What is new is their application to this model, which yields the dichotomy and the carrier classification above.

2 The surface, its source and the observation field

Throughout,

$$S(t, u) = (c + 2u(1 - c), (1 - u)s, uQ), \quad c = \cos t, \quad s = \sin t, \quad Q = \sqrt{c(2 - c)}, \quad (1)$$

on the source $D = [-\pi/2, \pi/2] \times [0, 1]$. Put $D_\circ = (-\pi/2, \pi/2) \times [0, 1]$ and $D_+ = [0, \pi/2] \times [0, 1]$. The edges and the polar ruling map to

$$S(\pm \frac{\pi}{2}, u) = (2u, \pm(1 - u), 0), \quad S(0, u) = (1, 0, u). \quad (2)$$

The two edge segments lie on distinct lines through the apex $(2, 0, 0)$. The ruling line $\{(1, 0, z)\}$ is skew to both edge lines.

Proposition 2.1 (Completed source and five rank labels [2]). *Near $t = \epsilon\pi/2$, $\epsilon \in \{1, -1\}$, let $r = Q \geq 0$, let $C(r) = 1 - \sqrt{1 - r^2}$, and put $B_\epsilon(r, u) = (C + 2u(1 - C), \epsilon(1 - u)\sqrt{1 - C^2}, ur)$. This extends analytically to $|r| < 1$ and agrees with S on the physical overlap. The rank-loss labels of the completed source are exactly*

$$P_0 = (0, 1), \quad P_\pm = (\pm t_0, u_0), \quad E_\pm = (\pm \pi/2, 0), \quad c_0 = \frac{3 - \sqrt{5}}{2}, \quad t_0 = \arccos c_0, \quad u_0 = \frac{1 - c_0}{2}.$$

Each is a cross-cap of the smooth extension. The adapted determinants are $\Delta(P_0) = 1$, $\Delta(P_\pm) = -5\sqrt{\sqrt{5} - 2}$ and $-\epsilon$ at $E_\pm$. The physical germs at P_0 and $E_\pm$ are a boundary half-germ and corner quarter-germs.

Sketch. $C(r)(2 - C(r)) = r^2$ gives $C(Q) = c$. Differentiation gives $(B_\epsilon)_r \times (B_\epsilon)_u = (\epsilon u, 2u, 0)$ at $r = 0$. On $c > 0$ the cross-product equations reduce to $c = 1, u = 1$, or to $u = (1 - c)/2 = c(2 - c)/2$, that is $c^2 - 3c + 1 = 0$. Since $S_{uu} = 0$, the cross-cap determinant reduces to $\det(S_u, S_{ut}, S_{tt})$, and the smooth recognition criterion [6, Cor. 4.5] applies. Full proofs, with Lean-checked rank statements, are in [2]. $\square$

The polynomial $c^2 - 3c + 1$ thus picks out the two lateral cross-caps. On $D_{\text{reg}} = D \setminus \{P_0, P_\pm, E_\pm\}$, a connected orientable surface with corners, $H_1(D_{\text{reg}}; \mathbb{Z}) = \mathbb{Z}^2$.

The observation field. On the angular chart $|t| < \pi/2$ the normal $V = S_t \times S_u$ has components $V_z = N$ and $V_x = M$, where

$$N = (1 - c)(1 - c - 2u), \quad M = \frac{c^2(2 - c) - u(1 - c + c^2)}{Q}, \quad F = N + iM, \quad \Psi = (N, M). \quad (3)$$

Both components are even in t . Since Q vanishes at $t = \pm\pi/2$, the field F and the planar map Ψ live on $D_\circ$. They depend on the chart, because reparametrizing the source multiplies V by its Jacobian.

3 Realizations of the surface by signal states

Let $S^3 = \{z \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}$, and let $C_0 = \{(0, w) : |w| = 1\}$ and $C_1 = \{(w, 0) : |w| = 1\}$. The moment map $z \mapsto (|z_1|^2, |z_2|^2)$ sends S^3 onto the segment $X + Y = 1$. To produce a surface one needs two further pieces of data: a choice of states over the source, and a map reading those states out.

Definition 3.1. Let $D' \subset D$. A *state family* on D' is any map $z : D' \rightarrow S^3$; continuity is not required. A *readout* is a map $P : U \rightarrow \mathbb{R}^3$ on a set $U \subset \mathbb{C}^2$ containing $z(D')$. The pair *realizes* S if $P \circ z = S$ on D' . The family is *moment-compatible* if $|z_1(t, u)|^2 = \cos t$, as in the original model. A readout is *real-analytic along* C_0 if $C_0 \subset U$ and $\beta \mapsto P(0, e^{i\beta})$ is real-analytic.

3.1 A realization with varying powers

Theorem 3.2 (Quadrature realization). *Put $\zeta_+(t) = e^{it/2}$ and $\zeta_-(t) = \sqrt{1 - c/2} + i\sqrt{c/2}$. The state family and the fixed real quadratic readout*

$$\Phi(t, u) = (\sqrt{1 - u}\zeta_+(t), \sqrt{u}\zeta_-(t)), \quad P(z) = (\Re z_1^2 + |z_2|^2 + \Re z_2^2, \Im z_1^2, \Im z_2^2)$$

satisfy $|\Phi_1|^2 = 1 - u$, $|\Phi_2|^2 = u$ and $P \circ \Phi = S$ on all of D . The phasors are analytic in the completed charts, and the family is smooth for $0 < u < 1$.

Proof. $\zeta_+^2 = c + is$ and $\zeta_-^2 = 1 - c + iQ$, so $P\Phi = ((1 - u)c + u(2 - c), (1 - u)s, uQ) = S$. In the edge chart $t = \epsilon \arccos C(r)$, which is analytic in r ; so is ζ_+ . The components of ζ_- become $\sqrt{1 - C/2}$ and $r/\sqrt{2(2 - C)}$, which are also analytic. $\square$

The phases encode an orthogonal-circle coupling. This is model data; power conservation alone does not select it.

Proposition 3.3 (Two limits of this realization). *(1) No C^1 state family can have $|z_2|^2 = u$ up to $u = 0$, or $|z_1|^2 = 1 - u$ up to $u = 1$. (2) No three fixed Hermitian 2×2 matrices H_j and normalized pure-state family satisfy $S_j = z^* H_j z$ on D .*

Proof. (1) At a zero of a C^1 component w , the derivative of $|w|^2$ is $2\Re(\bar{w}\partial w) = 0$, whereas $\partial_u u = 1$. (2) We have $z^* H_j z = a_j + (Mr(z))_j$, where $r(z)$ is the Bloch vector, a point of the unit sphere, and M is a real 3×3 matrix. If $\text{rank } M \leq 2$ the image is planar, but the values $(1, 0, 0), (0, 1, 0), (0, -1, 0), (1, 0, 1)$ of S have affine determinant 2. If M is invertible the image is an ellipsoid, which contains no segment, while $S(0, u) = (1, 0, u)$ is one [7]. $\square$

3.2 The original assignment

Under $|z_1|^2 = \cos t$, every state with $t = \pm\pi/2$ lies on C_0 , and every state with $t = 0$ lies on C_1 . Both edges are therefore read from a single circle of states, which a readout must map onto two different lines (Figure 2). This picture drives the rest of the section.

Proposition 3.4 (Phase-invariant and mixed-state readouts). *Suppose D' contains two distinct points of one edge $\{\pm\pi/2\} \times [0, 1]$. No readout with $P(e^{i\gamma}z) = P(z)$ realizes S on D' with a moment-compatible family. In particular no function of zz^* does, including any set of Hermitian expectation values. Likewise, for trace-one density matrices $\rho \geq 0$ with $\rho_{11} = \cos t$, no function of ρ realizes S .*

Proof. C_0 is a single orbit of the global phase, so such a readout is constant on C_0 , whereas S separates the two edge points. For density matrices, $|\rho_{12}|^2 \leq \rho_{11}\rho_{22} = 0$ on the edge, so $\rho = \text{diag}(0, 1)$ there. $\square$

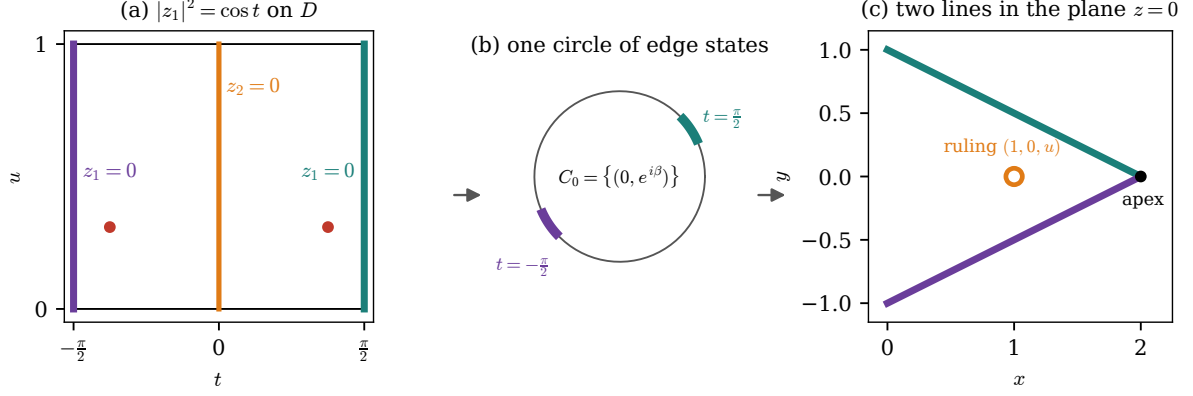


Figure 2: The mechanism behind Theorem 3.6. (a) Under the original powers, z_1 vanishes on both edges and z_2 on the polar ruling. (b) All edge states therefore lie on the circle C_0 ; the arcs shown are those used by the smooth readout of Theorem 3.8. (c) The edges map to a V in the plane $z = 0$, which no analytic image of C_0 can contain (Lemma 3.5). The ruling maps onto a segment over $(1, 0)$, on a line skew to both legs.

Lemma 3.5 (Analytic curves and lines). *Let $p : J \rightarrow \mathbb{R}^3$ be real-analytic on an open interval or on the circle. For every line ℓ , either $p(J) \subset \ell$ or $p(J) \cap \ell$ is countable. In particular $p(J)$ cannot contain uncountable subsets of two distinct lines.*

Proof. Choose $x_0 \in \ell$ and unit normals n_1, n_2 to ℓ . If $p(J) \not\subset \ell$, some $n_k \cdot (p - x_0)$ is a nonzero real-analytic function. Its zero set is discrete, hence countable, and it contains $p^{-1}(\ell)$. Two distinct lines meet in at most one point. $\square$

Theorem 3.6 (Analytic rigidity). *Let a moment-compatible family z and a readout P realize S on D' .*

1. *If $D' = D$, then P is not real-analytic along C_0 . More generally, P is not real-analytic on any connected open arc of C_0 containing all edge states $z(\pm\pi/2, u)$.*
2. *If $D' = D_\circ$, $C_0 \subset U$ and $P|_{U \cap S^3}$ is continuous at each point of C_0 , then P is not real-analytic along C_0 .*

Proof. (1) The image of C_0 , or of the arc, contains both edge segments of (2). These are uncountable subsets of distinct lines, contradicting Lemma 3.5. (2) Fix u and $\epsilon = \pm 1$, and take $t_n \rightarrow \epsilon\pi/2$. Since $|z_1(t_n, u)|^2 = \cos t_n \rightarrow 0$, a subsequence of $z(t_n, u) \in S^3$ converges to some $z^* \in C_0$. Continuity gives $P(z^*) = \lim S(t_n, u) = S(\epsilon\pi/2, u)$, so both edge segments lie in $P(C_0)$, and part (1) applies. $\square$

Corollary 3.7 (Excluded readouts). *Under the original assignment on D or $D_\circ$, S is not realized by any of the following:*

- an affine map $\mathbb{R}^4 \rightarrow \mathbb{R}^3$;
- a real polynomial map, or a polynomial in a bicomplex variable and its conjugates;
- a map that is real-analytic on a neighbourhood of C_0 .

On D this also excludes stereographic projection from a point of C_0 that avoids the edge states. In particular, the “geometric projection” of v12 Definition 3.1 cannot be taken from any of these classes.

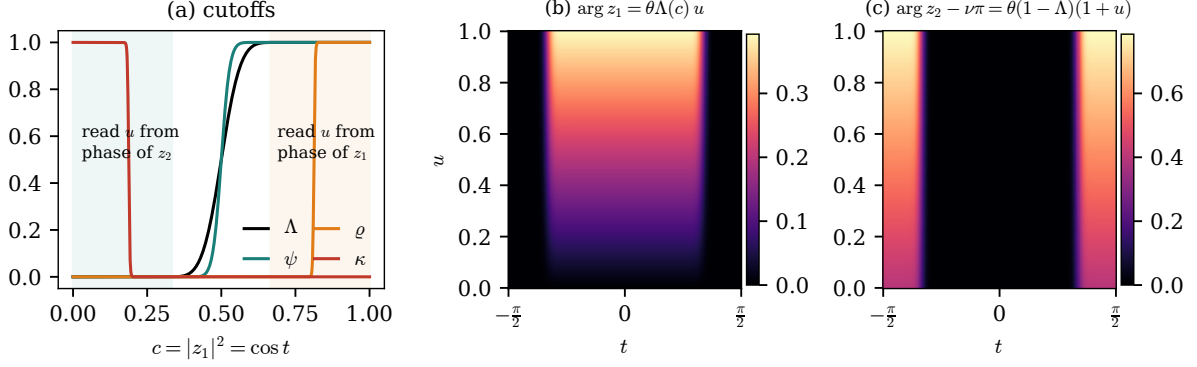


Figure 3: The switching readout of Theorem 3.8. (a) The cutoffs, as functions of $c = |z_1|^2$. Near the edges (c small) the ruling parameter u is stored in the phase of z_2 ; near the polar ruling (c close to 1) it is stored in the phase of z_1 . (b), (c) The two phases over the source; each is monotone in u wherever its decoder is used.

3.3 Sharpness

The obstruction is analytic rather than topological: smooth readouts exist. The construction below reads the ruling parameter u from a phase. Near the edges $c = 0$ it reads u from the phase of z_2 ; near the polar ruling $c = 1$ it reads u from the phase of z_1 . Smooth cutoffs in $c = |z_1|^2$ switch between these decoders (Figure 3). Table 1 records why every decoder is smooth wherever it is used.

Theorem 3.8 (C^∞ realization). *Put $\theta = \pi/8$. Take smooth functions $\Lambda, \psi, \varrho, \kappa : [0, 1] \rightarrow [0, 1]$ such that:*

- $\Lambda = 0$ on $[0, \frac{1}{4}]$, $\Lambda = 1$ on $[\frac{3}{4}, 1]$, and Λ is strictly increasing in between;
- $\psi = 0$ on $[0, \frac{1}{3}]$ and $\psi = 1$ on $[\frac{2}{3}, 1]$;
- $\varrho = 0$ on $[0, \frac{3}{4}]$ and $\varrho = 1$ on $[\frac{7}{8}, 1]$;
- $\kappa = 1$ on $[0, \frac{1}{8}]$ and $\kappa = 0$ on $[\frac{1}{4}, 1]$.

Let χ be a smooth 2π -periodic function equal to 1 on $[-\frac{\pi}{8}, \frac{3\pi}{8}]$ and to -1 on $[\frac{7\pi}{8}, \frac{11\pi}{8}]$, and let $\nu = 0$ for $t \geq 0$ and $\nu = 1$ for $t < 0$. Define

$$z(t, u) = (\sqrt{c} e^{i\theta\Lambda(c)u}, \sqrt{1-c} e^{i(\nu\pi + \theta(1-\Lambda(c))(1+u))}).$$

This is a moment-compatible family. It is continuous on D , smooth on $D_\circ$, and smooth up to $t = \pm\pi/2$ in the edge charts, since there $\sqrt{C(r)} = r/\sqrt{1 + \sqrt{1-r^2}}$. With $c = |z_1|^2$ and principal arguments, set

$$\begin{aligned} \mathcal{U} &= \psi(c) \frac{\arg z_1}{\theta\Lambda(c)} + (1 - \psi(c)) \left(\frac{\arg z_2^2}{2\theta(1 - \Lambda(c))} - 1 \right), \\ \Sigma &= \varrho(c) \Re z_2 \sqrt{1+c} + (1 - \varrho(c)) |z_2| \chi(\arg z_2) \sqrt{1+c}, \\ \mathcal{Q} &= \kappa(c) \Re z_1 \sqrt{2-c} + (1 - \kappa(c)) \sqrt{c(2-c)}, \end{aligned}$$

where each summand is read as zero where its cutoff vanishes. Then $P = (|z_1|^2 + 2\mathcal{U}|z_2|^2, (1 - \mathcal{U})\Sigma, \mathcal{U}\mathcal{Q})$ is smooth on an open neighbourhood of $z(D)$, and $P \circ z = S$ on D . Hence S has a C^∞ readout defined on all of $\mathbb{C}^2$.

Decoder	Used where	Why it is smooth there	Value on the states
$\arg z_1/(\theta\Lambda)$	$\psi > 0$, so $c > \frac{1}{3}$	$\Lambda > 0$ and $z_1 \neq 0$; $\arg z_1 \in [0, \theta]$	u
$\arg z_2^2/(2\theta(1-\Lambda)) - 1$	$\psi < 1$, so $c < \frac{2}{3}$	$\Lambda < 1$ and $ z_2 ^2 > \frac{1}{3}$; $\arg z_2^2 \in [0, \frac{\pi}{2}]$	u
$\Re z_2 \sqrt{1+c}$	$\varrho > 0$, so $c > \frac{3}{4}$	smooth; here $\Lambda = 1$, so z_2 is real	s
$ z_2 \chi(\arg z_2) \sqrt{1+c}$	$\varrho < 1$, so $c < \frac{7}{8}$	$ z_2 ^2 > \frac{1}{8}$; $\arg z_2$ lies in a plateau of χ	s
$\Re z_1 \sqrt{2-c}$	$\kappa > 0$, so $c < \frac{1}{4}$	smooth; here $\Lambda = 0$, so $z_1 = \sqrt{c}$	Q
$\sqrt{c(2-c)}$	$\kappa < 1$, so $c > \frac{1}{8}$	$0 < c < 2$	Q

Table 1: The six decoders of Theorem 3.8. Wherever a cutoff is positive, its decoder is smooth on a neighbourhood of the states there and takes the listed value.

Proof of Theorem 3.8. For $c \geq \frac{3}{4}$ we have $\Lambda = 1$, so $z_2 = \operatorname{sgn}(t)\sqrt{1-\cos t} = \sqrt{2}\sin(t/2)$, which is smooth; at $t = \pm\pi/2$ only $\sqrt{c} \rightarrow 0$ enters. On the states, $\arg z_1 = \theta\Lambda u$, $\arg z_2^2 = 2\theta(1-\Lambda)(1+u)$, and $\arg z_2 \in \nu\pi + [0, \pi/4]$. Table 1 follows directly: each decoder takes the stated value wherever its cutoff is nonzero. In the third and fourth rows, $\Re z_2 \sqrt{1+c} = \operatorname{sgn}(t)\sqrt{1-c^2} = s$ and $\chi(\arg z_2) = \operatorname{sgn} t$.

For smoothness, each summand is defined on the union of two open sets. On the first its decoder is smooth. On the second its cutoff, a smooth function of $|z_1|^2$, vanishes identically. The two definitions agree on the overlap, and the union contains $z(D)$. Hence P is smooth on an open set $U \supset z(D)$, and substitution gives $P \circ z = S$. Since $z(D)$ is compact, choose an open V with $z(D) \subset V$ and $\bar{V} \subset U$ compact, and a smooth bump equal to 1 on V and supported in U . Its product with P is a C^∞ readout on $\mathbb{C}^2$. $\square$

On C_0 this readout traces the two legs of the V on the two disjoint arcs of Figure 2(b), and the plateau function χ switches between them. This is exactly the flexibility that analytic readouts lack.

Lemma 3.9 (Analytic retraction). *Let M be a real-analytic surface without boundary, $z : M \rightarrow \mathbb{R}^4$ a real-analytic immersion, and $K \subset M$ a compact set on which z is injective. Then there are an open $M' \supset K$ on which z is an embedding, an open tube $T \supset z(K)$, and a real-analytic map $\varpi : T \rightarrow M'$ with $\varpi \circ z = \operatorname{id}$ near K .*

Proof. If z were not injective on any neighbourhood of K , there would be $x_n \neq y_n$ with $z(x_n) = z(y_n)$ and both sequences accumulating on K . Any limits x, y satisfy $z(x) = z(y)$, so $x = y$, which contradicts local injectivity of an immersion near x . Shrink to a relatively compact M' , on which z is then an embedding onto $\Sigma = z(M')$. Gram-Schmidt applied to an analytic local frame gives analytic normal frames, so the normal bundle of Σ is real-analytic. The map $E(x, n) = x + n$ is real-analytic with invertible differential along the zero section, hence a local analytic diffeomorphism there [8]. Compactness gives injectivity on $\{|n| < \delta\}$ over a neighbourhood of $z(K)$. Put $\varpi = z^{-1} \circ \operatorname{pr} \circ E^{-1}$. $\square$

Proposition 3.10 (One half of the source). *(1) If D' contains at least three points of the ruling $\{0\} \times [0, 1]$ and at least three points of one edge, no affine readout realizes S on D' with a moment-compatible family. This applies in particular to D_+ , which contains the single interior cross-cap P_+ . (2) On D_+ there are a real-analytic moment-compatible family and a readout, real-analytic on a neighbourhood of its image, that realize S .*

Proof. (1) Write $P = b + L$. On C_1 the readout traces an ellipse centred at b , possibly degenerate, through three collinear points of the ruling line. A nondegenerate ellipse meets a line in at most

two points, so the ellipse is a segment and b lies on the ruling line. The same argument on C_0 puts b on the edge line, but the two lines are skew.

(2) The closed half source has corners, so we work on an open analytic surface M containing it. M is glued from the angle chart $(-\delta, \pi/2) \times (-\delta, 1 + \delta)$ and the edge chart $(-\delta, \delta) \times (-\delta, 1 + \delta)$ in (r, u) ; there S extends analytically. Fix $0 < \theta < \pi$ and put $z = e^{i\theta u}v$ with $v = (\sqrt{\cos t}, \sqrt{2} \sin(t/2))$; in the edge chart $v = (r/\sqrt{1 + \sqrt{1 - r^2}}, (1 - r^2)^{1/4})$. Both expressions are analytic on M , and $|z_1|^2 = \cos t$ on D_+ . Since $\partial_u z = i\theta z$ and $\partial_t z = e^{i\theta u}v'$ with v' real and nonzero, z is an immersion. It is injective on D_+ : if $e^{i\theta u}v = e^{i\theta u'}v'$ with v, v' nonnegative, the phase quotient is positive, so $u = u'$ and $t = t'$. Lemma 3.9 with $K = D_+$ gives ϖ , and $S \circ \varpi$ is the required readout. $\square$

Remark 3.11 (An edge criterion). Lemma 3.5 gives a simple test for any power assignment $w = |z_1|^2$. An analytic readout needs $S(w^{-1}(0))$ to lie in the image of an analytic loop, and a phase-invariant readout needs it to be a single point. The same holds for $1 - w$.

The assignment of Theorem 3.2 passes this test. Its vanishing edges $u = 1$ and $u = 0$ map to arcs of the circles $(x - 1)^2 + z^2 = 1$ and $x^2 + y^2 = 1$, and its readout maps the two vanishing circles onto exactly these circles. The original assignment fails, because $w^{-1}(0)$ consists of both edges.

A realization of the original model therefore needs one ingredient beyond the moment map:

- a non-analytic switching readout, as in Theorem 3.8;
- a half source, which keeps one cross-cap and loses the reflection pairing $S(t, 1) = S(-t, 1)$; or
- different powers.

Polynomial readouts can approximate S uniformly to any accuracy, by the Weierstrass theorem applied to Theorem 3.8, but never exactly. Whether a quadratic readout exists on D_+ is an interesting open question.

4 Observation monodromy and its carriers

4.1 Interior dipole and square-root restoration

Theorem 4.1 (Dipole and two-traversal restoration). *The zeros of F in $D_\circ$ are $P_\pm$ and the boundary point $(0, 1)$. With the (t, u) orientation [2],*

$$\det DF(P_\pm) = \mp \frac{5s_0(1 - c_0)^2}{Q_0}, \quad \text{Ind}(P_\pm) = \mp 1.$$

Every continuous branch of $\sqrt{F}$ transported round a small positive loop about either point changes sign; two traversals restore it.

Proof. $N = 0$ exactly on $c = 1$ and on $u = (1 - c)/2$. On the first, $M = 1 - u$. On the second, M is a positive multiple of $-(c^2 - 3c + 1)$. The determinant follows from $c_0^2 = 3c_0 - 1$, and a nondegenerate zero has small-circle winding equal to the sign of its determinant. Write $F(\gamma(s)) = \exp L(s)$ with a continuous logarithm, so that $L(1) = L(0) + 2\pi i n$ with $n = \pm 1$. Then $\chi = \exp(L/2)$ satisfies $\chi(1) = -\chi(0)$, and any other continuous root differs from χ by a constant sign. $\square$

The restoration is topological: it counts traversals. Proposition 4.3 identifies the structure that carries it. Two numerical values are worth separating. The Jacobian modulus is $|\det DF(P_\pm)| \approx 2.245$. The value $\sqrt{5}$, by contrast, is $|P'(c_0)|$ for the monic polynomial $P(c) = c^2 - 3c + 1$. Both N and M are even in t ; the opposite signs of the two determinants come from the t -derivatives.

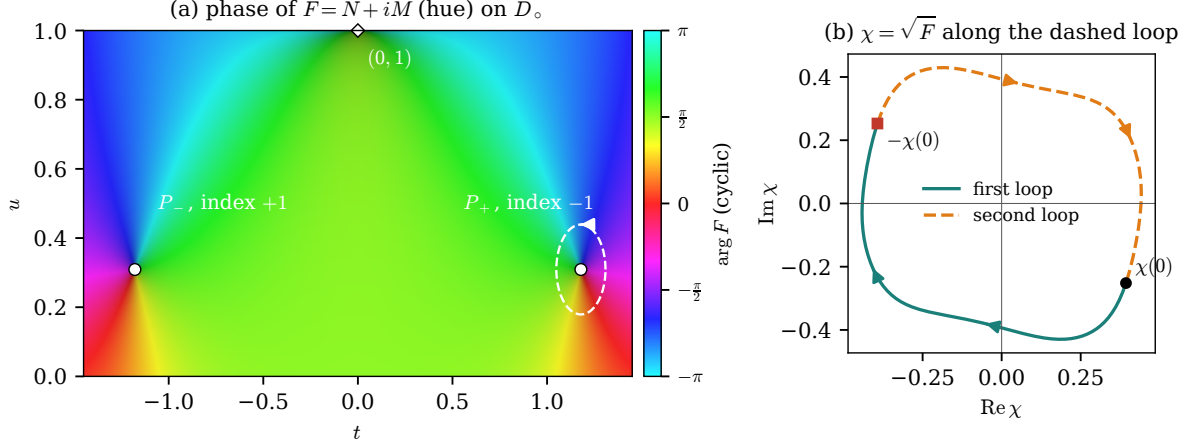


Figure 4: (a) The phase of the observation field F on $D_\circ$, shown for $|t| \leq 1.45$, with the cyclic colour scale on the right; brightness increases with $|F|$. Every colour meets $P_\pm$, the dipole of Theorem 4.1. (b) A continuous square root χ of F along the positively oriented dashed loop about P_+ , with arrows in the direction of travel: one traversal ends at $-\chi(0)$, two traversals restore $\chi(0)$.

4.2 The polar fold

Proposition 4.2 (Polar fold). *The smooth extension of Ψ at $P_0 = (0, 1)$ is a Whitney fold of local degree zero.*

Proof. Use the source coordinates $a = -M$ and $b = 2 \sin(t/2) \sqrt{u - (1 - \cos t)/2}$, and the target coordinates $(N, M) \mapsto (-M, -N)$. The derivative matrix at $(0, 1)$ is invertible, and the exact identity $-N = b^2$ gives the normal form (a, b^2) . Reflecting b pairs preimages of opposite orientation. $\square$

The exact charts make the conclusion independent of Taylor remainders; the remainder displayed in v12 omitted the term $M(t, 1) = -t^4/2 + O(t^6)$. The full interior critical locus of Ψ consists of two parts [3, Thm. 2.4]: the symmetry segment $t = 0$, which is an ordinary fold for $u > 0$; and a rational branch, which consists of ordinary folds away from $(0, 0)$, where the ordinary-fold condition fails (Figure 5).

4.3 Three sign structures

Proposition 4.3 (Carriers). *(1) For connected X , $\text{End}_{D(X)}(\mathbb{Z}_X) = H^0(X; \mathbb{Z}) = \mathbb{Z}$. Every involution of $\mathbb{Z}_X$ is therefore $\pm \text{id}$, $\mathbb{Z}_X$ admits no splitting into two nonzero summands, and its characteristic polynomial is $t - 1$ or $t + 1$. The same holds for the constant condensed sheaf on a compact connected space, whose global sections are the continuous maps to $\mathbb{Z}$.*

(2) Let $X = D_\circ \setminus \{P_+, P_-, (0, 1)\}$ and let $\pi : \tilde{X} \rightarrow X$ be the double cover $\{\chi^2 = F\}$. Then $\pi_ \mathbb{Z}$ is a rank-two local system with deck involution τ , indecomposable over $\mathbb{Z}$. After inverting 2, $\pi_* \mathbb{Z}[\frac{1}{2}] \simeq \mathbb{Z}[\frac{1}{2}] \oplus \mathcal{L}[\frac{1}{2}]$, where the sign system $\mathcal{L}$ has monodromy -1 around each of $P_\pm$.*

(3) The fold involution $(\xi, \eta) \mapsto (\xi, -\eta)$ and the deck involution τ cannot be identified: the first fixes the fold arc, the second is free.

(4) For a standard cross-cap image X' with topological normalization ν , the rational normalization defect $\mathcal{Q}_\nu = \nu_ \mathbb{Q}_{\tilde{X}'} / \mathbb{Q}_{X'}$ is $j_! \mathcal{L}_{\text{sgn}}$, the extension by zero of the sign line on the open double ray. Its stalk at the pinch is zero, so every morphism from $\mathcal{Q}_\nu$ to a skyscraper at the pinch vanishes.*

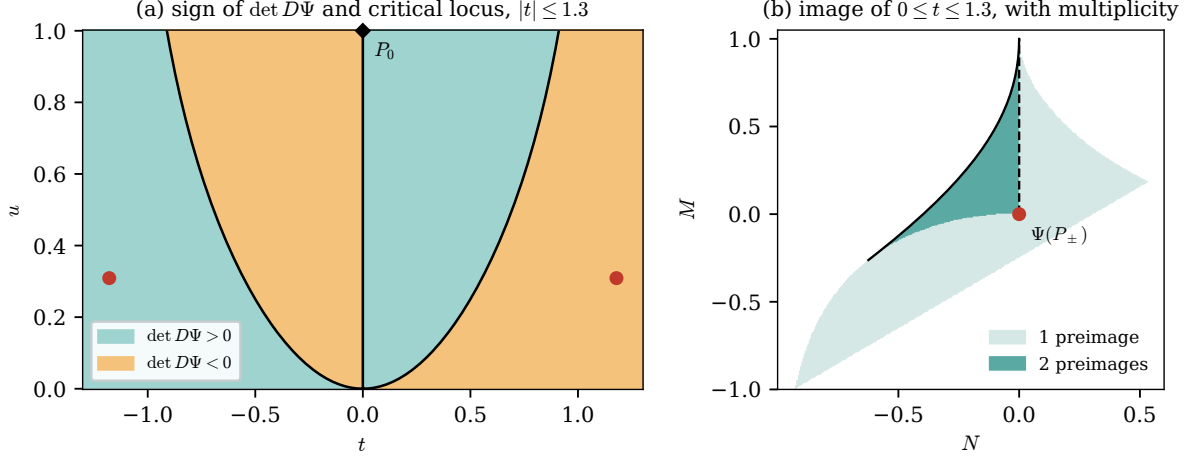


Figure 5: The planar observation map $\Psi = (N, M)$. (a) The sign of $\det D\Psi$ on $|t| \leq 1.3$ and the critical locus: the symmetry segment $t = 0$ and the rational branch through $(0, 0)$. (b) Since N and M are even in t , $\Psi(t, u) = \Psi(-t, u)$, so it suffices to draw the image of $0 \leq t \leq 1.3$. The shading counts preimages in this half. The solid curve is the image of the rational branch, the dashed segment that of $t = 0$, and both lateral cross-caps map to the origin. On $|t| \leq 1.3$ every count doubles: for instance $(N, M) = (-0.1, 0.2)$ has the four preimages $(\pm 0.9947, 0.3375)$ and $(\pm 0.3687, 0.7778)$.

Proof. (1) Degree-zero endomorphisms of a sheaf in degree zero are the global sections of the sheaf $\mathcal{H}om(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}$, and a direct-sum splitting corresponds to an idempotent. The only idempotents of $\mathbb{Z}$ are 0 and 1.

(2) $D_\circ \setminus \{(0, 1)\}$ is star-shaped about an interior point, so $\pi_1(X)$ is free on loops around $P_\pm$, on which F winds ∓ 1 . The monodromy of π is the surjection onto C_2 , and the stalk of $\pi_*\mathbb{Z}$ is the regular module $\mathbb{Z}[C_2] = \mathbb{Z}[x]/(x^2 - 1)$. Local systems correspond to π_1 -modules, so

$$\mathrm{End}(\pi_*\mathbb{Z}) \simeq \mathrm{End}_{\pi_1(X)}(\mathbb{Z}[C_2]) = \mathrm{End}_{\mathbb{Z}[C_2]}(\mathbb{Z}[C_2]) \simeq \mathbb{Z}[C_2],$$

and direct-sum decompositions correspond to idempotents. If $(a + bx)^2 = a + bx$, then $b(2a - 1) = 0$, so $b = 0$ and $a \in \{0, 1\}$. Over $\mathbb{Z}[\frac{1}{2}]$ the projectors $(1 \pm x)/2$ split the module.

(3) The fixed-point sets differ.

(4) Near the cross-cap, ν is the proper source map, with one preimage over the pinch and off the double ray, and two preimages over the open double ray. Proper base change gives $(\nu_*\mathbb{Q})_p = \mathbb{Q}^{\nu^{-1}(p)}$. At one-preimage points $\mathbb{Q} \rightarrow \mathbb{Q}$ is the identity, so the quotient stalk is 0. At two-preimage points the diagonal $\mathbb{Q} \rightarrow \mathbb{Q}^2$ has quotient $\mathbb{Q}$, through $(a, b) \mapsto a - b$, and exchanging the sheets acts on it by -1 . These stalk maps are compatible with restriction. So $\mathcal{Q}_\nu$ restricts to the anti-invariant line on the ray and vanishes at the pinch, which identifies it with $j_!\mathcal{L}_{\mathrm{sgn}}$ [12]. Finally, $\mathrm{Hom}(\mathcal{Q}_\nu, i_*\mathbb{Q}) = \mathrm{Hom}(i^*\mathcal{Q}_\nu, \mathbb{Q}) = 0$. $\square$

The three structures are thus cleanly separated. The sign system $\mathcal{L}$ lives on the punctured source and records the winding of F ; the normalization line lives on the image double stratum and records the exchange of sheets; and the constant sheaf carries no sign at all. Relating $\mathcal{L}$ to the normalization line would require a new identification of these two $\mathbb{Z}/2$ -torsors, which would be a natural object to construct.

4.4 Exceptional pullbacks and folds

We use the six-functor formalism on finite-dimensional locally compact Hausdorff spaces with $D(X) = D(\mathrm{Sh}(X; \mathbb{Z}))$ [9, Ex. 5.9], $\omega_X = a_X^!\mathbb{Z}$ and $\mathbb{D}_X = \mathrm{RHom}(-, \omega_X)$. Three standard facts

are used:

- Verdier duality, $f^! \mathbb{D}_Y \simeq \mathbb{D}_X f^*$ [10, Prop. 4.6.9], [11, Ch. III];
- the stalk of ω_X at x is the local homology $H_*(X, X \setminus x)$, and $\omega_X \simeq \text{or}_X[n]$ on an n -manifold [10, Rem. 4.6.19, Cor. 4.6.20];
- open immersions satisfy $j^! = j^*$ and are cohomologically smooth [9, Rem. 5.4].

A map f is cohomologically smooth when the comparison $f^! \mathbb{Z} \otimes f^*(-) \rightarrow f^!(-)$ is an isomorphism, $f^! \mathbb{Z}$ is invertible, and both properties persist under base change [9, Def. 5.1].

Theorem 4.4 ($f^! \mathbb{Z}$ depends only on the domain). (1) For any continuous $f : X \rightarrow Y$ between topological manifolds without boundary of dimensions n and m , $f^! \mathbb{Z}_Y \simeq \text{or}_X \otimes f^* \text{or}_Y[n - m]$.

(2) Let $D' \subset \mathbb{R}^2$ be locally closed and convex with nonempty interior, and let $j : \text{int } D' \rightarrow D'$ be the inclusion. For every continuous $\Psi : D' \rightarrow \mathbb{R}^2$, $\Psi^! \mathbb{Z} \simeq j_! \mathbb{Z}_{\text{int } D'}$.

Proof. (1) Since ω_Y is invertible, $\mathbb{Z}_Y \simeq \mathbb{D}_Y \omega_Y$, so $f^! \mathbb{Z}_Y \simeq \mathbb{D}_X f^* \omega_Y = \text{RHom}(f^* \text{or}_Y[m], \text{or}_X[n])$.

(2) If x is not an interior point, then a point strictly between an interior point and any point of D' is interior. So $D' \setminus x$ is star-shaped, hence contractible, and the local homology at x vanishes. At interior points it is $\mathbb{Z}[2]$. Therefore $\omega_{D'} \simeq j_! \mathbb{Z}_{\text{int } D'}[2]$, and

$$\Psi^! \mathbb{Z}_{\mathbb{R}^2} \simeq \Psi^! \mathbb{D}_{\mathbb{R}^2}(\mathbb{Z}_{\mathbb{R}^2}[2]) \simeq \mathbb{D}_{D'} \Psi^*(\mathbb{Z}_{\mathbb{R}^2}[2]) = \mathbb{D}_{D'}(\mathbb{Z}_{D'}[2]) = \omega_{D'}[-2]. \quad \square$$

Corollary 4.5. For Ψ of (3) on $D_\circ$ and any inclusion $i : L \rightarrow D_\circ$, the complex $i^* \Psi^! \mathbb{Z}$ is $\mathbb{Z}$ in degree 0 on $L \cap \text{int } D_\circ$ and 0 on the sides $u = 0, 1$. In particular the formula $i^* \Psi^! \mathbb{Z} \simeq \mathbb{Z}[-1]$ stated for the fold locus in the earlier public description holds at no point. For an embedded open arc L in the interior, $i^! \Psi^! \mathbb{Z} \simeq \text{or}_L[-1]$, a shift that comes from the codimension of L .

Proposition 4.6 (Where the fold is visible). Let x be an interior ordinary fold point of Ψ , and choose disc neighbourhoods $V \ni x$ and $W \ni \Psi(x)$ on which $f = \Psi|_V$ is homeomorphic to $(\xi, \eta) \mapsto (\xi, \eta^2)$. Let L be the fold arc, let $W' \subset W$ be the open side of the discriminant containing $f(V \setminus L)$, and put $G = j_!^* \mathbb{Z}_{W'}$. Then $f^! \mathbb{Z}_W \simeq \mathbb{Z}_V$, $f^* G \simeq (\mathbb{Z}_{V \setminus L})_!$ and $f^! G \simeq \mathbb{Z}_V$. The comparison map $f^! \mathbb{Z}_W \otimes f^* G \rightarrow f^! G$ is injective with cokernel $i_* \mathbb{Z}_L$. So at every ordinary fold point invertibility of $f^! \mathbb{Z}$ holds, and the comparison condition is exactly what fails.

Proof. The first isomorphism is Theorem 4.4(1). Pullback commutes with extension by zero and $f^{-1}(W') = V \setminus L$, which gives the second. For the closed side $k : H \rightarrow W$, local homology vanishes on its edge, so $\mathbb{D}_W(\mathbb{Z}_H) = k_* \omega_H \simeq G[2]$. Since $f(V) \subset H$, we get $f^! G[2] \simeq \mathbb{D}_V(f^* \mathbb{Z}_H) = \mathbb{D}_V(\mathbb{Z}_V) = \mathbb{Z}_V[2]$.

Let $\phi : (\mathbb{Z}_{V \setminus L})_! \rightarrow \mathbb{Z}_V$ be the comparison map, read through these identifications, and compute its kernel $\mathcal{K}$ and cokernel $\mathcal{C}$ stalk by stalk. Off L , f is locally an open immersion, so ϕ is an isomorphism there, and $\mathcal{K}_y = \mathcal{C}_y = 0$. On L the source stalk is 0 and the target stalk is $\mathbb{Z}$, so $\mathcal{K}_y = 0$ and $\mathcal{C}_y = \mathbb{Z}$. Thus ϕ is injective and $\mathcal{C}$ is supported on L . The surjection $\mathbb{Z}_V \rightarrow \mathcal{C}$ restricts to a surjection $\mathbb{Z}_L \rightarrow i^* \mathcal{C}$, which is an isomorphism stalkwise, since a surjection $\mathbb{Z} \rightarrow \mathbb{Z}$ is bijective. Hence $\mathcal{C} \simeq i_* \mathbb{Z}_L$. $\square$

The fold also shows up in $f_* \mathbb{Z}_V$, whose stalk ranks are 2, 1, 0 across the discriminant, and on the surface side in the normalization defect. A compact Hausdorff image defines a condensed set. The singularity information, however, comes from the maps above rather than from the condensed structure itself (v12 Remark 8.7).

5 Residual geometry

5.1 Sublevel law and resolution cost

For the standard models take Lebesgue measure and the residuals $K_f = x^2 + y^4$ and $K_\times = x^2(1 + y^2) + y^4$.

Theorem 5.1 (Sublevel law). *With $B = B(\frac{1}{4}, \frac{3}{2})$ and $\varepsilon > 0$,*

$$V_f(\varepsilon) = B\varepsilon^{3/4}, \quad B\varepsilon^{3/4}(1 + \sqrt{\varepsilon})^{-1/2} \leq V_\times(\varepsilon) \leq B\varepsilon^{3/4}.$$

The exponent $3/4$ is unchanged by smooth positive densities and by smooth changes of source or target coordinates fixing the origin; the coefficient may change.

Proof. Slicing in y gives $4 \int_0^{\varepsilon^{1/4}} \sqrt{\varepsilon - y^4} dy$. Substituting $y = \varepsilon^{1/4}v$ and then $z = v^4$ yields $B\varepsilon^{3/4}$. For the cross-cap, the extra factor $(1 + y^2)^{-1/2}$ lies between $(1 + \sqrt{\varepsilon})^{-1/2}$ and 1.

The sublevel sets shrink to the origin, so a smooth positive density or the Jacobian of a source diffeomorphism fixing 0 is a positive constant plus $o(1)$ on them, and V is multiplied by a factor tending to a positive constant. A target diffeomorphism fixing 0 changes the squared residual by a factor between positive constants $a \leq b$, so $V(\varepsilon/b) \leq V'(\varepsilon) \leq V(\varepsilon/a)$. Neither change affects the exponent. $\square$

The geometric resolution cost is therefore $-\log V(\varepsilon) = \frac{3}{4} \log \varepsilon^{-1} - \log B + o(1)$, a clean, parameter-free law. A communication capacity is a different quantity: it depends on a channel. For example, identical binary labels admit a constant channel of capacity 0 and a noiseless one of capacity $\log 2$. Accordingly the conjecture $C = \sqrt{5} \ln S$ from the earlier description is replaced by this resolution law.

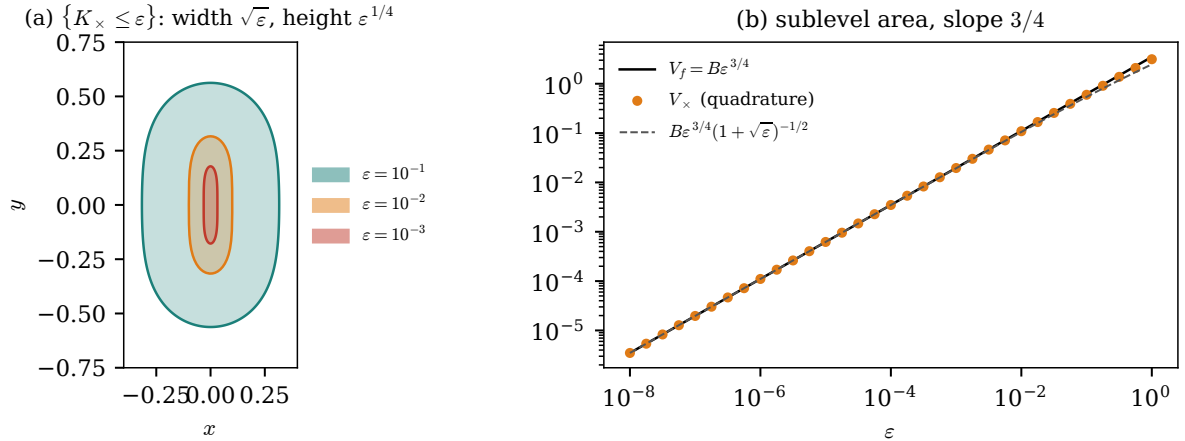


Figure 6: The sublevel law of Theorem 5.1. (a) The sets $\{K_\times \leq \varepsilon\}$ shrink anisotropically, with width $\sqrt{\varepsilon}$ and height $\varepsilon^{1/4}$, so their area scales as $\varepsilon^{1/2+1/4}$. (b) The cross-cap area, computed by quadrature, between the two bounds of the theorem.

5.2 An explicitly chosen complex completion

Theorem 5.2 (Chosen A_3 completion). *After its coefficients are extended to $\mathbb{C}$, the germ of $K_\times$ is right-equivalent to $X^2 + Y^4$, with Milnor and Tjurina numbers 3. The fiber $U = \{X^2 + Y^4 = 1\}$ is a smooth genus-one curve E with two points removed. $H^1(U)$ has dimension three and monodromy spectrum $\{-i, 1, i\}$. The weight-graded pieces of its mixed Hodge structure have dimension two in weight one and one in weight two.*

Proof. The substitution $X = x\sqrt{1+y^2}$, $Y = y$ is a holomorphic coordinate change with invertible derivative. The Milnor algebra is $\mathbb{C}[X, Y]/(2X, 4Y^3)$, of dimension 3. By Euler's identity $X^2 + Y^4 = \frac{X}{2} \cdot 2X + \frac{Y}{4} \cdot 4Y^3$ lies in the Jacobian ideal, so the Tjurina number is also 3. The weighted compactification $X^2 + Y^4 = Z^4 \subset \mathbb{P}(2, 1, 1)$ is a double cover of $\mathbb{P}^1$ branched at four points, hence of genus one, and its two ends at infinity are unramified. Weighted homogeneity identifies the global and local Milnor fibers [13, 14]. The monodromy $h(X, Y) = (-X, iY)$ acts by $-i$ and i on $H^1(E)$ and fixes the boundary class. The Gysin sequence $0 \rightarrow H^1(E) \rightarrow H^1(U) \rightarrow \tilde{H}^0(\{\infty_{\pm}\})(-1) \rightarrow 0$ gives the weights [15, 16]. $\square$

These Hodge-theoretic data belong to the chosen complexification. A full nearby-cycle package would additionally need the specialization and variation maps, which the monodromy alone does not determine.

5.3 Three different projected curves

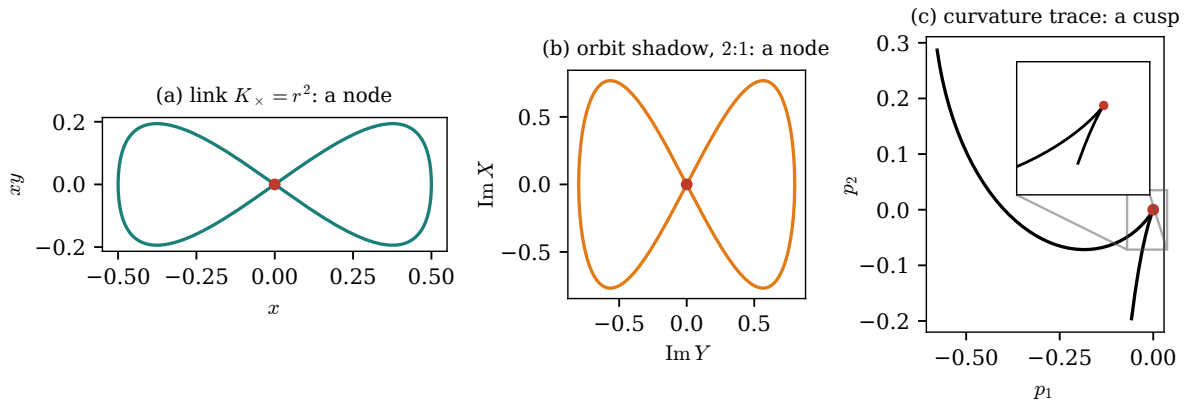


Figure 7: The three curves of Proposition 5.3. (a) The image of the real link $K_{\times} = r^2$ ($r = \frac{1}{2}$), projected to the (x, xy) -plane; the node is the double point $(0, 0, r)$. (b) The shadow $(\Im Y, \Im X)$ of a weighted orbit ($b = 0.8$). (c) The pedal trace $p = (p_1, p_2)$ of the exceptional principal curvature at the lateral cross-cap P_+ , computed from the exact invariants. The inset magnifies the ordinary cusp at the origin, where both branches share one tangent.

Proposition 5.3. (1) For each $r > 0$ the real link $\{K_{\times} = r^2\}$ is a circle whose cross-cap image has one transverse double point, a figure-eight graph. The lift (x, xy, y^2, y) is an embedding. (2) The weights of $X^2 + Y^4$ are $(2, 1)$. For $a, b > 0$ with $a^2 + b^4 = 1$, the orbit $\Gamma(s) = (ae^{2is}, be^{is})$ is embedded, and its shadow $(b \sin s, a \sin 2s)$ is a transverse figure-eight with frequency ratio $2 : 1$. The action $2\pi(4a^2 + b^2)$ has infimum 2π over nondegenerate orbits, attained only in their closure; $b^2 = \frac{1}{8}$ is a strict maximum. (3) For the actual lateral germs the Euclidean Whitney invariants satisfy $b_{\pm}^2 + a_{\pm}c_{\pm}^2 = 0$, and the exceptional principal-curvature trace has an ordinary cusp at its double return.

Proof. (1) The gradient of $K_{\times}$ vanishes only at 0. Equal images force $x = x'$ and $y' = \pm y$, and the tangent images at $(0, \pm\sqrt{r})$ are transverse.

(2) The Y coordinate fixes the phase. With $z = b^2$, the action divided by 2π is $1 + (1 - z)(4z + 3)$.

(3) Use the adapted pair $\xi = \partial_u/\sqrt{3}$ and $\eta = -\partial_t + \lambda_{\pm}\partial_u$, where $\lambda_{\pm} = -c_0 s_{\pm}/[2(1 - c_0)]$. Let Π be the projection onto the normal plane of S_u , and put $A = \Pi(S_{tt} - 2\lambda_{\pm}S_{tu})$ and $B = \Pi(-S_{tu}/\sqrt{3})$. The actual second jet gives

$$|A|^2 = \frac{-335+162\sqrt{5}}{12}, \quad \langle A, B \rangle^2 = \frac{-520+249\sqrt{5}}{27}, \quad R = |B|^2 - \frac{\langle A, B \rangle^2}{|A|^2} = \frac{20(140+11\sqrt{5})}{11397} > 0.$$

Oriented Gram–Schmidt then gives

$$a = -\frac{\langle A, B \rangle^2}{2|A|^3}, \quad b_{\pm} = \pm \frac{\sqrt{\langle A, B \rangle^2 R}}{|A|^2}, \quad c = \frac{\sqrt{2R}}{|A|^{1/2}},$$

so $b^2 + ac^2 = 0$ identically. In the Euclidean normal form the pedal trace is

$$p(\tau) = \frac{2(c\tau^2 + 2b\tau - ac)}{c^2 + 4\tau^2} (-c, 2\tau)$$

[17, Prop. 2.1, Eq. (2.8)]. Its numerator equals $c(\tau - \tau_*)^2$ with $\tau_* = -b/c$, and

$$\det(p''(\tau_*), p'''(\tau_*)) = -\frac{96c^3}{(c^2 + 4b^2/c^2)^2} \neq 0,$$

which is the ordinary-cusp criterion. □

The three constructions produce three different curves, a node, a node and a cusp, and each comes with its own data (Figure 7).

6 Open problems

- Does a quadratic readout realize S on D_+ under the original powers (Remark 3.11)? Is there a natural principle that selects a non-analytic readout?
- Construct a sign-twisted, differential-form or spinor operator whose boundary data see the carriers of Proposition 4.3, and relate the phase sign system to the normalization line.
- The singular Dirac and Pin problem of v12 Sections 8.20–8.24. This requires closed domains for the twisted Dirac operator of the actual Whitney metric, graph and overlap estimates, singular-cut trace spaces, and Clifford-compatible transmission maps. The exact-cone and regular-cut results are the model cases. The scalar theory that serves as the first step is developed in [4].
- Build signal or physical models in which capacity, mass, clocks or chirality are defined independently, and compare them with the geometry above.

A Review of the earlier record

A.1 The public description

The description field of the record [1] shows the abstract of an earlier draft, which differs from both public files. Its claims compare with the present results as follows:

Claim in the description	Present status
S is derived from the moment map and the power constraint	Not by analytic readouts (Theorem 3.6); smooth realizations exist (Theorem 3.8)
Immersion singularities occur at exactly two lateral points	Five rank labels (Proposition 2.1)
Quasi-separated condensed set; Weil-étale/Pontryagin connection	Replaced by Bohr–Haar orthogonality (§B) and Proposition 4.3(1)
$i^*\Psi^! \mathbb{Z} \simeq \mathbb{Z}[-1]$ on the fold locus	$\Psi^! \mathbb{Z} \simeq j_! \mathbb{Z}$ (Corollary 4.5); folds seen by Proposition 4.6
Monodromy as an endomorphism of the constant condensed sheaf	Carried by $\pi_* \mathbb{Z}$ (Proposition 4.3)
$T > 2(M-1)/\delta$ gives independence	Independence for all $T > 0$ (§B)
$C = \sqrt{5} \ln S$ (conjectural)	Replaced by the 3/4 resolution law (§5)

A.2 v11 Theorem 4.2

v11 identified the sign reversal of $\sqrt{F}$ with an exchange of the sheets of a Whitney fold; v12 dropped that identification, and Proposition 4.3(3) shows why. In all other blocks up to 8.20, v11 agrees with v12 apart from small refinements of wording. v11 stated the spinorial parametrix as a remark, which matches the present open problem.

A.3 All named statements of v12

Categories: **H** holds, possibly with an explicit domain or hypothesis; **C** holds after correction; **S** strengthened; **R** does not hold as defined; **O** open; **M** model-level or conditional; **D** missing definition, settled here. “C.” refers to the companion paper [4], which proves the operator statements for general surfaces with Whitney cross-caps; Section 9.2 of C. treats the present surface.

Block	Cat.	Note and location
3.1	D	Analytic readouts excluded; smooth readout exists; other powers admit a quadratic readout (§3)
3.2	C	Fix $L^2(dx)$ and the conjugate-linear first slot; the introduction used another measure (§B)
3.3	H	Geometric resolution cost (§5)
4.1	C	F is an angular-chart field on $D_\circ$; indices ∓ 1 (§2, Thm 4.1)
4.2	H	Two-traversal sign restoration (Thm 4.1)
4.3	H	Topological statement, as v12 says
4.4	C	N and M are both even; corrected parity explanation (§4)
5.1	H	$ P'(c_0) = \sqrt{5}$ for the monic P
5.2	C	The lateral Jacobian modulus is about 2.245; its sign follows the orientation
5.3	H	Exact fold constant; two-sided cross-cap bounds (Thm 5.1)
5.4	H	Uses an external identification $S = \varepsilon^{-1}$
5.5	H	Capacity depends on a channel
5.6	H	Explicit holomorphic coordinate change (Thm 5.2)
5.7	H	Genus one, two unramified ends, spectrum $\{-i, 1, i\}$
5.8	H	Every $r > 0$ for the standard model (Prop. 5.3)
5.9	H	The orbit leaves the fixed fiber except at fourth-root phases
5.10	C	Infimum attained only in the closure; $b^2 = \frac{1}{8}$ is a maximum
5.11	H	Scope remark
5.12	H	Invariants and cusp criterion (Prop. 5.3)
5.13	H	Geometric no-go
6.1	C	$c^2 - 3c + 1$ selects the lateral labels; five rank labels (Prop. 2.1)
6.2	C	Five cross-cap germs, including two corner quarter-germs
7.1	C	Formal adjoint parameter depends on the measure; domains made explicit
7.2	H	Indices $(1, 0)$ with explicit H_0^1 and H^1 domains
7.3	C	For $\sigma > \frac{1}{2}$ the functions are L^2 with nonzero trace
7.4	H	Conventions made explicit
7.5	H	For a fixed finite family
7.6	S	Independence for every $T > 0$; the threshold controls conditioning
7.7	H	Bohr torus realization
7.8	H	Conditioning and log-determinant bounds
8.1	H	Full-line and half-line realizations distinguished
8.2	H	Evaluation boundary $\sigma = \frac{1}{2}$
8.3	H	Two separate thresholds
8.4	H	Character orthogonality
8.5	H	Lattices of lower rank retained
8.6	H	Normalization defect (Prop. 4.3(4))
8.7	H	Sharpened by Thm 4.4
8.8	C	Fold proved through exact charts; remainder term added (Prop. 4.2)

Block	Cat.	Note and location
8.9	C	All five labels removed; $H_1 = \mathbb{Z}^2$
8.10	H	On finite-energy conforming spaces, $J_n^* G_{n+1} J_n = G_n$ and $J_n^* K_{n+1} J_n = K_n$ follow from sesquilinearity, since $\varphi_i^{(n)} = \sum_{\alpha} (J_n)_{\alpha i} \varphi_{\alpha}^{(n+1)}$; no convergence is claimed
8.11	C	Closures are form closures. Zero capacity holds at $P_{\pm}$ (C. Prop. 4.1), and at $P_0, E_{\pm}$ by restricting the same cutoffs to the physical germ
8.12	O	The model identities hold; the uniform estimate for the actual germ is unproved. It is not needed: arclength coordinates make the metric uniformly elliptic (C. Lemma 3.2)
8.13	R	The graph closure of the interior core has infinite deficiency from the outer boundary (C. Prop. 8.1). The corrected operator $H_F _{\ker \tau}$ has indices (2, 2) and the stated logarithmic boundary form (C. Thms. 5.2, 6.1, 6.3)
8.14	R	Fails for the old minimum, which has a Neumann extension (C. Prop. 8.1); holds for the corrected operator (C. Cors. 7.1, 7.2)
8.15	R	Compactness and the gap hold (C. Thm. 4.2). The Krein kernel of the old minimum is infinite-dimensional (C. Prop. 8.1); that of the corrected operator is two-dimensional (C. Thm. 5.2)
8.16	H	For the actual H_0 : strictly positive, with Friedrichs extension H_F , discrete reduced Krein spectrum and the weak buckling equivalence (C. Prop. 8.1)
8.17	C	Operator scope: the fixed-boundary Friedrichs operator and its point restriction (C. §§4–7)
8.18	H	Support and parity obstruction (Prop. 4.3)
8.19	H	Scope remark
8.20	M	Exact-cone model (§6)
8.21	O	Actual front–seam parametrix (§6)
8.22	M	First-order data hold; a Pin convention must be chosen
8.23	M	Conditional on a Clifford-compatible transmission
8.24	H	Open-problem statement
8.25	H	For a specified proper lift (§C)
8.26	H	Representation observation
8.27	H	Real-geometry Deligne/Weil scope
8.28	H	Weights of the chosen completion (Thm 5.2)
8.29	H	Monodromy versus the full nearby-cycle package
8.30	H	For a specified Weil realization
9.1	H	The ratio is a length
9.2	H	Numerical pattern
9.3	H	Power coordinates Poisson-commute
9.4	H	A separate observable programme
9.5	H	$\eta = 0$ for the sign circle
9.6	H	Sign cover and spectral symmetry

B Spectral and arithmetic estimates

The earlier versions used the following classical facts. We collect them in the precise form the model uses; they are independent of the realization and monodromy results of the main text.

Mellin operators. Let $T_{\sigma} f = -i(xf' + \sigma f)$. For C^1 functions on $[a, b] \subset (0, \infty)$, $\int_a^b (\overline{T_{\sigma} f} g - \overline{f} T_{1-\sigma} g) dx = i[x\overline{f}g]_a^b$. In $L^2(dx)$ the formal adjoint is therefore $T_{1-\sigma}$, which equals T_{σ} exactly at $\sigma = \frac{1}{2}$; for the measure $x^{2\beta-1} dx$ the parameter becomes $2\beta - \sigma$. The map $(Uf)(s) = e^{s/2} f(e^s)$ is unitary from $L^2([1, \infty), dx)$ onto $L^2([0, \infty), ds)$. It carries the closure of $T_{1/2}$ on C_c^{∞} to $-i\partial_s$ on H_0^1 , whose adjoint has domain H^1 . The deficiency indices are (1, 0), so the half-line operator has no self-adjoint extension, while on the full multiplicative line it is self-adjoint [18]. The functions $x^{-\sigma-i\omega}$ satisfy $T_{\sigma} x^{-\sigma-i\omega} = -\omega x^{-\sigma-i\omega}$. They are generalized eigenfunctions at $\sigma = \frac{1}{2}$, and for $\sigma > \frac{1}{2}$ they are square-integrable but have nonzero trace at $x = 1$.

Finite Gram matrices. For distinct real $\omega_1, \dots, \omega_M$, $T > 0$ and $G_{jk} = T^{-1} \int_0^T e^{i(\omega_j - \omega_k)t} dt$, the phasors are linearly independent and G is positive definite for *every* $T > 0$. Indeed, a vanishing combination and its first $M - 1$ derivatives at an interior time give a Vandermonde system. With $\varepsilon = 2(M - 1)/(T \min |\omega_j - \omega_k|) < 1$, the spectrum of G lies in $[1 - \varepsilon, 1 + \varepsilon]$, $\kappa_2(G) \leq (1 + \varepsilon)/(1 - \varepsilon)$ and $M \log(1 - \varepsilon) \leq \log \det G \leq 0$. The threshold $T > 2(M - 1)/\delta$ of the earlier versions therefore quantifies conditioning.

Hardy evaluation and Bohr–Haar. On the coefficient Hilbert space of Dirichlet series, evaluation at $\Re s = \sigma$ is bounded exactly when $\sigma > \frac{1}{2}$, with norm $\zeta(2\sigma)^{1/2}$ [19]. The Gram limit $\lim T^{-1} \int_0^T n^{it} m^{-it} dt = \delta_{nm}$ is character orthogonality on the product torus of prime exponents. That torus also records relations: a finite set of integers may generate a lattice of lower rank, as $\{2, 4\}$ does.

Proposition B.1 (Log-scale and linear-scale windows). *Let $K(\delta, L) = L^{-1} \int_0^L e^{-i\delta\tau} d\tau$, so that $|K| = |\sin(\delta L/2)|/(|\delta|L/2) \leq 2/(|\delta|L)$, and let $T > 1$.*

1. $\frac{1}{\log T} \int_1^T x^{-i\delta} \frac{dx}{x} = K(\delta, \log T)$. *The log-scale and linear-scale bounds are the single bound $2/(|\delta| \cdot \text{mass})$ evaluated at window masses $\log T$ and T , for every real δ .*
2. *With Lebesgue measure on the same interval, $\frac{1}{T-1} \int_1^T x^{-i\delta} dx = \frac{T^{1-i\delta}-1}{(1-i\delta)(T-1)}$, whose modulus tends to $(1 + \delta^2)^{-1/2}$.*
3. *For $e^{-i\delta t}$ on $[1, T]$ with Haar measure, the average is at most $2/(|\delta| \log T)$ in modulus.*

Proof. Substitute $x = e^\tau$; integrate directly; integrate by parts once. □

So the two rates compare window masses, T against $\log T$, and the characters of $(\mathbb{R}_{>0}, \times)$ decorrelate under Haar measure but not under Lebesgue measure. This replaces the integer-sparsity explanation given in the earlier description; note also that the positive integers generate the dense subgroup $\mathbb{Q}_{>0}$.

C Scope statements carried over

The following statements of v12 hold as stated there:

- A nonempty proper subset of $SU(2)$ is not invariant under the full left action. This applies to a constrained source lift once one is specified.
- The ratio $2\pi^2 R^3/(\pi r^2)$ with $r = \phi^{-2}R$ equals $2\pi\phi^4 R$. It is a length, so a mass law would need its own scale and operator.
- For the canonical Kähler form the two power coordinates Poisson-commute. The full $SU(2)$ moment map generates a noncommutative algebra and is a natural starting point for quantization.
- The antiperiodic circle operator $-i\partial_\theta$ has the symmetric spectrum $\mathbb{Z} + \frac{1}{2}$ and vanishing eta invariant [20].
- The pair (E, D) of Theorem 5.2 can be evaluated by any Weil cohomology theory with localization and Tate twists, giving a one-dimensional boundary term; such a theory is an additional choice.

D Evidence and reproducibility

The historical authority is the public v12 file, which is byte-identical to the archived copy. The SHA-256 values of the two public files are

```
v12  4ebb8f0a7d6e88a6c5cd97f65d224b4e938867cabad54f057d3ceafb24167d5a
v11  207a51feed62fe7f5a6dd7c9190258dd1b9c996c8b2735e799aff569bb017ec2
```

All proofs are given in full in the text. The figures are generated by a script from the explicit formulas of the paper; their numerical panels illustrate results proved in the text.

Exact and finite replay. Independent scripts check the following:

- the derivative determinants, the Euclidean jet invariants of Proposition 5.3(3) and the polar remainder;
- the quadrature and Bloch identities;
- the edge and skew-line algebra;
- a replay of the C^∞ readout on 801×81 source labels, which reproduces S to 1.4×10^{-14} ;
- the analytic edge-chart identities on the half source;
- the Jacobian factorization, the windings ∓ 1 and the idempotents of $\mathbb{Z}[C_2]$;
- the window kernels.

They confirm the algebra the written proofs rely on.

Software companion. The verifier scripts, the figure generator with its hash manifest, the claim map and the Lean project below are archived as a software companion [5], with DOI [10.5281/zenodo.23103299](https://doi.org/10.5281/zenodo.23103299).

Lean. A companion Lean project (Lean 4, Mathlib) contains 33 named theorems, audited for proof holes and custom axioms. They include:

- the actual rank classification, the endpoint ranks and the polar fold;
- the small-circle indices and the evenness of the field;
- the monic derivative;
- the injectivity and fibres of the cross-cap lift;
- square-root and logarithm lifting;
- the Mellin interval identity, and the Gram entries with their eigenvalue enclosure;
- the quadrature readout identities and the cusp factorization.

This is partial coverage of selected statements.

External theorems. The following external results are used, each with its hypotheses verified for the actual maps:

- smooth cross-cap recognition [6];
- the real-analytic inverse function theorem [8];
- Milnor fiber comparison and mixed Hodge theory [13, 14, 15, 16];
- the Euclidean pedal formula [17];
- Verdier duality and six-functor smoothness [10, 9, 11];
- Dirichlet–Hardy space facts [19];
- self-adjointness by Fourier transform [18].

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